Question 1 — Science

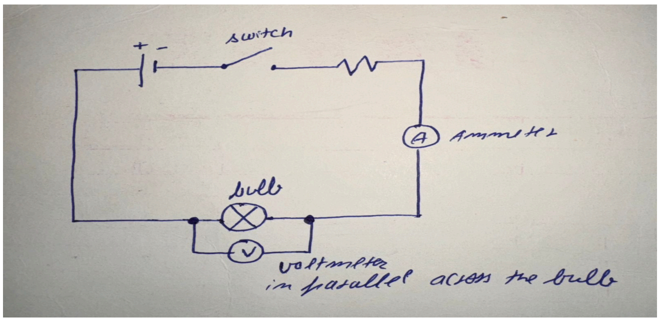
A simple electric circuit is a closed conducting path through which electric current can flow. The main components are a battery, a switch, a bulb, a resistor, an ammeter and a voltmeter.

The battery is the source of energy. It provides the potential difference that drives the current around the circuit. The switch is used to open or close the circuit. When the switch is closed the path is complete and current flows through the resistor and the bulb, so the bulb glows. When the switch is open the path is broken and no current flows, so the bulb does not glow.

The battery, switch, resistor, bulb and ammeter are all connected in series in the main circuit. The ammeter is connected in series because it measures the current passing through the circuit, and the same current flows through every component in a series circuit. The voltmeter is connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb.

The conventional current flows from the positive terminal of the battery, through the switch, resistor, ammeter and bulb, and back to the negative terminal. I have marked this direction with an arrow in my diagram.

The current depends on both the potential difference and the resistance. According to Ohm's law, $V = IR$. If the voltage of the battery stays the same and the resistance is increased, the current in the circuit decreases and the bulb becomes dimmer. If the resistance is reduced, more current flows and the bulb becomes brighter.



Question 2 — English

Technology has made information far easier to reach, but easier access does not by itself make a student a better learner. What matters is what the student does with the access. Used well, technology removes real obstacles: a student who did not follow an explanation in class can watch three different explanations at home until one makes sense, and a student who cannot afford textbooks can use a digital library. In my own experience, watching a topic explained a second way has often been the thing that made it click.

The risk is that the same access lets a student skip the difficulty entirely. Looking up the answer to every hard question means the work gets finished without the thinking that the work was meant to produce. Someone who copies the steps of a maths solution can reproduce them and still fail a slightly different question, because they never worked out why those steps were chosen.

So technology does not make students better or worse learners on its own. It makes both outcomes easier to reach. It helps most when it is used to understand something or to check your own reasoning, and it harms most when it is used to avoid reasoning at all. It should support thinking rather than replace it.

Question 3 — Economics

(a) The graph is drawn below with price on the vertical axis and quantity on the horizontal axis. The demand curve slopes downward from left to right, because consumers buy a greater quantity when the price is lower. The supply curve slopes upward from left to right, because producers are willing to supply a greater quantity when the price is higher.

(b) The two curves intersect at a price of ₹30 and a quantity of 60 units. This point is the market equilibrium, because at this price the quantity demanded is exactly equal to the quantity supplied, so there is no pressure on the price to change.

(c) If the market price is below ₹30, the quantity demanded is greater than the quantity supplied. This creates a shortage, because consumers want to buy more than producers are willing to sell, and the shortage pushes the price upward. If the market price is above ₹30, the quantity supplied is greater than the quantity demanded. This creates a surplus, because producers have more goods than consumers want to buy, and the surplus pushes the price downward.

(d) If the cost of production increases, producing the good becomes less profitable at every existing price, so producers are willing to supply less at each price. The supply curve shifts to the left. I have shown the new supply curve as S_1 in my graph.

Assuming demand is unchanged, the new equilibrium occurs at a higher price and a lower quantity than before.